6. Suppose we collect data for a group of students in a statistics class with variables X_1 = hours studied, X_2 = undergrad GPA, and Y = receive an A. We fit a logistic regression and produce estimated coefficient, $\hat{\beta}_0 = -6, \hat{\beta}_1 = 0.05, \hat{\beta}_2 = 1$.

(a) Estimate the probability that a student who studies for 40 h and has an undergrad GPA of 3.5 gets an A in the class.

X_1 = hour studied

X_2 = undergrad GPA

$$p = \sigma(\mu) = \frac{1}{1+e^{-\mu}} \qquad \mu = \overset{-6}{\beta_0} + \overset{0.05}{\beta_1} X_1 + \overset{1}{\beta_2} X_2$$

(a) $X_1 = 40$

$X_2 = 3.5$

$$\mu = -6 + 0.05 \times 40 + 1 \times 3.5 = -0.5$$

$$p = \sigma(0.5) = \frac{1}{1+e^{-0.5}} = \frac{1}{1+1.6487} \approx 0.3775 = 37.8\% \#$$

(b) How many hours would the student in part (a) need to study to have a 50 % chance of getting an A in the class?

$$p = \sigma(\mu) = \frac{1}{1+e^{\mu}} = 0.5 \qquad X_2 = 3.5$$

$$1 + e^{\mu} = 2 \qquad e^{\mu} = 1 \qquad \mu = 0$$

$$0 = -6 + 0.05 \times X_1 + 1 \times 3.5$$

X_1 = 50 hour #

9. This problem has to do with *odds*.

(a) On average, what fraction of people with an odds of 0.37 of defaulting on their credit card payment will in fact default?

$$\text{odds} = \frac{p}{1-p}$$

(a)

$$0.37 = \frac{p}{1-p}$$

$$0.37(1-p) = p$$

$$p = \frac{0.37}{1.37} \quad p \approx 0.27 = 27\% \ \#$$

(b) Suppose that an individual has a 16 % chance of defaulting on her credit card payment. What are the odds that she will default?

$p = 0.16$

$$\text{odds} = \frac{0.16}{1-0.16} = \frac{0.16}{0.84} \approx 0.19 \ \#$$

12. Suppose that you wish to classify an observation $X \in \mathbb{R}$ into apples and oranges. You fit a logistic regression model and find that

$$\widehat{\Pr}(Y = \texttt{orange}|X = x) = \frac{\exp(\hat{\beta}_0 + \hat{\beta}_1 x)}{1 + \exp(\hat{\beta}_0 + \hat{\beta}_1 x)}.$$

Your friend fits a logistic regression model to the same data using the *softmax* formulation in (4.13), and finds that

$$\widehat{\Pr}(Y = \texttt{orange}|X = x) = \frac{\exp(\hat{\alpha}_{\texttt{orange}0} + \hat{\alpha}_{\texttt{orange}1} x)}{\exp(\hat{\alpha}_{\texttt{orange}0} + \hat{\alpha}_{\texttt{orange}1} x) + \exp(\hat{\alpha}_{\texttt{apple}0} + \hat{\alpha}_{\texttt{apple}1} x)}.$$

(a) What is the log odds of orange versus apple in your model?

(a)

$$\text{odds} = \frac{\widehat{Pr}(Y = orange \mid X = x)}{\widehat{Pr}(Y = apple \mid X = x)} = \frac{\widehat{Pr}(Y = orange \mid X = x)}{1 - \widehat{Pr}(Y = orange \mid X = x)}$$

$$\mu = \hat{\beta}_0 + \hat{\beta}_1 x$$

$$= \frac{\hat{p}(x)}{1 - \hat{p}(x)} = \frac{\frac{e^{\mu}}{1+e^{\mu}}}{1 - \frac{e^{\mu}}{1+e^{\mu}}} = \frac{\frac{e^{\mu}}{1+e^{\mu}}}{\frac{1}{1+e^{\mu}}} = e^{\mu}$$

$$\log \frac{\hat{p}(x)}{1 - \hat{p}(x)} = \log e^{\mu} = \mu = \hat{\beta}_0 + \hat{\beta}_1 x$$

(b) What is the log odds of orange versus apple in your friend's model?

$$\text{odds} = \frac{p}{1-p} = \frac{\frac{e^{(\hat{\alpha}_{00} + \hat{\alpha}_{01} x)}}{e^{(\hat{\alpha}_{00} + \hat{\alpha}_{01} x)} + e^{(\hat{\alpha}_{a0} + \hat{\alpha}_{a1} x)}}}{1 - \frac{e^{(\hat{\alpha}_{00} + \hat{\alpha}_{01} x)}}{e^{(\hat{\alpha}_{00} + \hat{\alpha}_{01} x)} + e^{(\hat{\alpha}_{a0} + \hat{\alpha}_{a1} x)}}}$$

$$= \frac{e^{(\hat{\alpha}_{00} + \hat{\alpha}_{01} x)}}{e^{(\hat{\alpha}_{a0} + \hat{\alpha}_{a1} x)}} = e^{(\hat{\alpha}_{00} - \hat{\alpha}_{a0}) + (\hat{\alpha}_{01} - \hat{\alpha}_{a1}) x}$$

$$\log \frac{p}{1-p} = (\hat{\alpha}_{00} - \hat{\alpha}_{a0}) + (\hat{\alpha}_{01} - \hat{\alpha}_{a1}) x$$

(c) Suppose that in your model, $\hat{\beta}_0 = 2$ and $\hat{\beta}_1 = -1$. What are the coefficient estimates in your friend's model? Be as specific as possible.

$$(\overbrace{\hat{\alpha}_{00} - \hat{\alpha}_{a0}}^{2}) + (\overbrace{\hat{\alpha}_{01} - \hat{\alpha}_{a1}}^{-1}) x = \hat{\beta}_0 + \hat{\beta}_1 x$$

$$\hat{\alpha}_{00} = 2 + C_0, \quad \hat{\alpha}_{a0} = C_0, \quad \hat{\alpha}_{01} = -1 + C_1, \quad \hat{\alpha}_{a1} = C_1$$

$\Rightarrow$ not only one solution

(d) Now suppose that you and your friend fit the same two models on a different data set. This time, your friend gets the coefficient estimates $\hat{\alpha}_{\text{orange}0} = 1.2$, $\hat{\alpha}_{\text{orange}1} = -2$, $\hat{\alpha}_{\text{orange}0} = 3$, $\hat{\alpha}_{\text{orange}1} = 0.6$. What are the coefficient estimates in your model?

$$\hat{\beta}_0 = \hat{\alpha}_{00} - \hat{\alpha}_{a0} = 1.2 - 3 = -1.8$$

$$\hat{\beta}_1 = \hat{\alpha}_{01} - \hat{\alpha}_{a1} = -2 - 0.6 = -2.6$$

(e) Finally, suppose you apply both models from (d) to a data set with 2,000 test observations. What fraction of the time do you expect the predicted class labels from your model to agree with those from your friend's model? Explain your answer.

100%. Both models make identical predictions for every test observation because in d, the softmax parameters imply the same log-odds $\beta_0 = \alpha_{00} - \alpha_{a0}$, $\beta_1 = \alpha_{01} - \alpha_{a1}$. Hence they produce the same probabilities.